The 15:30 forecast's weights, and when sign(s) makes money

866 deck days, 2020-01-03 .. 2024-04-30; forecast of record: per-bar ridge, live-feasible features, 2000-session window, bar ending 16:00

Question. Which inputs move the 15:30 variance forecast, how stable are their weights, and do the weights or the inputs' contributions line up with the days on which sign(s) (buy the nearest-OTM 0DTE package if $s = r\hat{v} - iv > 0$, sell otherwise, settle at the close) makes money?

Verification (before any reading). The fitted coefficients are not stored, so the forecast was re-run with the research code and the coefficient vector recorded at every refit. (1) The re-run equals the stored forecast on all 1,469 rows of the 16:00 bar (max relative difference 7.3×10^{-12}). (2) Coefficients · inputs + intercept reproduce every forecast to 2.1×10^{-15}, and forecast = level + $\sum_j$ contribution$_j$ to 1.2×10^{-15}, where contribution$_j = \beta_j(x_j - \bar{x}_j)$ over the trailing window and the level is the window-mean target (to 4.9×10^{-13}). (3) All 866 deck days join (bar-end 16:00 ET row = forecast issued at 15:30); 52 are month-end sessions. Standardized weight = $\beta_j \times$ trailing-window sd of x_j. Intervals: circular block bootstrap, block $\lceil n^{1/3} \rceil$, 2,000 draws; * = 95% interval excludes 0.

Money on this forecast. Summed return per premium: mid +102.98 (month-ends −10.67, other days +113.65); crossed (buy at the ask, sell at the bid) +74.15 (−12.94 / +87.09). It buys 348 of 866 days and only 10 of the 52 month-ends.

Table 1: The 12 inputs that move the forecast most on the deck days (by sd of their contribution; forecast sd 0.357 in the same units).

input	mean std. weight	sd	days > 0	contribution sd
realized variance (target), last bar	+0.068	0.018	100%	0.066
4th-power returns (tails), mean of last 625 bars	-0.028	0.013	6%	0.045
realized variance (target), mean of last 5 bars	+0.048	0.009	100%	0.044
realized variance (target), mean of last 25 bars	+0.047	0.007	100%	0.041
absolute return, last bar	+0.043	0.004	100%	0.040
signed return, mean of last 5 bars	-0.038	0.009	0%	0.040
bipower variation, mean of last 625 bars	-0.024	0.002	0%	0.035
realized variance (target), mean of last 125 bars	+0.039	0.005	100%	0.033
absolute return, mean of last 5 bars	+0.033	0.006	100%	0.032
upside squared returns, mean of last 625 bars	-0.020	0.002	0%	0.031
absolute return, mean of last 25 bars	+0.030	0.004	100%	0.029
VIX, last bar	+0.017	0.013	100%	0.027

Table 2: Series contribution vs the day's return per premium (the top 8 series by contribution sd, the FOMC-day flag, and the forecast itself).

	all (866)		month-ends (52)		other (814)	
series	mid	crossed	mid	crossed	mid	crossed
realized variance (target)	-0.02	-0.02	-0.07	-0.07	-0.01	-0.01
absolute return	-0.05*	-0.06*	-0.06	-0.06	-0.04	-0.05
4th-power returns (tails)	+0.05	+0.06	-0.04	-0.04	+0.06	+0.07
signed return	+0.01	+0.01	-0.08	-0.08	+0.02	+0.02
VIX	+0.00	-0.00	+0.04	+0.04	+0.00	-0.00
VIX3M	-0.03	-0.03	-0.11	-0.11	-0.03	-0.03
ES volume	+0.02	+0.01	-0.15	-0.15	+0.04	+0.04
bipower variation	+0.01	+0.01	-0.15	-0.16	+0.03	+0.03
FOMC day flag	-0.02	-0.02	+0.20*	+0.20*	-0.03	-0.04
forecast (sum)	-0.01	-0.02	-0.09	-0.10	+0.00	-0.00

Feature level (138 inputs): intervals excluding 0, standardized weight / contribution (chance ≈ 7 of 138)

split	mid P&L	crossed P&L	buy decision
all	1 / 14	0 / 16	37 / 21
month-end	4 / 13	4 / 16	1 / 7
other	0 / 5	0 / 7	38 / 27

Table 3: What drives the decision: correlation of the two sides of s with the day's mid return and with the buy decision (+1 buy, −1 sell). $\log \hat{rv}$ = log diurnal scale (not a regression output) + the regression part.

	all		month-ends		other	
	mid	decision	mid	decision	mid	decision
log forecast $\hat{rv}$	-0.05	-0.02	-0.02	+0.03	-0.05	-0.02
log implied variance	-0.06	-0.25*	-0.07	-0.20	-0.05	-0.25*
$\log(\hat{rv}/iv)$ (sign = decision)	+0.04	+0.76*	+0.17	+0.70*	+0.02	+0.76*
log diurnal scale of $\hat{rv}$	-0.05	-0.05	+0.02	+0.13	-0.06	-0.06
regression part of $\log \hat{rv}$	-0.00	+0.04	-0.05	-0.13	+0.01	+0.06
log diurnal scale − log implied	+0.02	+0.33*	+0.12	+0.43*	-0.00	+0.31*

Reading.

- *The weights keep their sign and move in steps.* The penalty is re-chosen every 250 refits (1000, the top of the grid, on 712 of 866 deck days; 100 on the rest), and the inputs that move the forecast most keep one sign on every deck day: realized variance in the last bar (+0.068, sd 0.018), its 5- and 25-bar means (+0.048, +0.047), absolute return in the last bar (+0.043), the 5-bar signed return (−0.038). The forecast is a HAR on realized variance and absolute returns (series contribution sd 0.170 and 0.111), trimmed by long-window tail, bipower and upside terms (negative weights), with VIX adding and VIX3M subtracting (0.038 / 0.037).
- *Weights do not track money.* Across 138 inputs, 1 standardized-weight interval excludes zero for the mid return on all days (0 crossed; 0 on non-month-end days) – below the ≈7 expected by chance. They do co-move with the buy/sell mix (37 of 138 exclude zero for the decision): both drift slowly over the sample, which is not a day-level link to the return.
- *Contributions barely track money.* 14 (mid) and 16 (crossed) of 138 contribution intervals exclude zero on all days, against ≈7 by chance; on non-month-end days 5 and 7, i.e. chance. At the series level only absolute return is outside zero on all days (−0.049 mid, −0.058 crossed): days on which recent absolute returns push the forecast up earn less. The forecast itself: −0.014 [−0.07, +0.04]. No quintile pattern is monotone (Figure 3).
- *The decision is set by the implied side and the diurnal scale, not by the weights.* $\log(\hat{rv}/iv)$ has variance 0.101 while each side has about 1 (correlation 0.952); of $\log \hat{rv}$'s variance 0.967, the diurnal scale carries 0.813 and the regression part 0.365 (covariance −0.211). The regression part correlates +0.04 with the decision (interval through zero); log implied variance −0.25 on non-month-end days. Of the 330 day-to-day decision changes, the forecast moved more than the implied on 134.
- *Month-ends.* This forecast sells 42 of the 52 month-ends and loses there. On month-ends the FOMC-day flag's contribution correlates +0.20 with the day's return (interval +0.09 .. +0.35); on the 10 month-end buys (4 winners, 6 losers) the winners had lower absolute-return and ES-volume contributions and higher tail contributions. Both rest on a handful of days.

Files: `experiments/model_diagnostics_1530.py` (capture, then analyze); outputs in `results/model_diagnostics_1530/`.

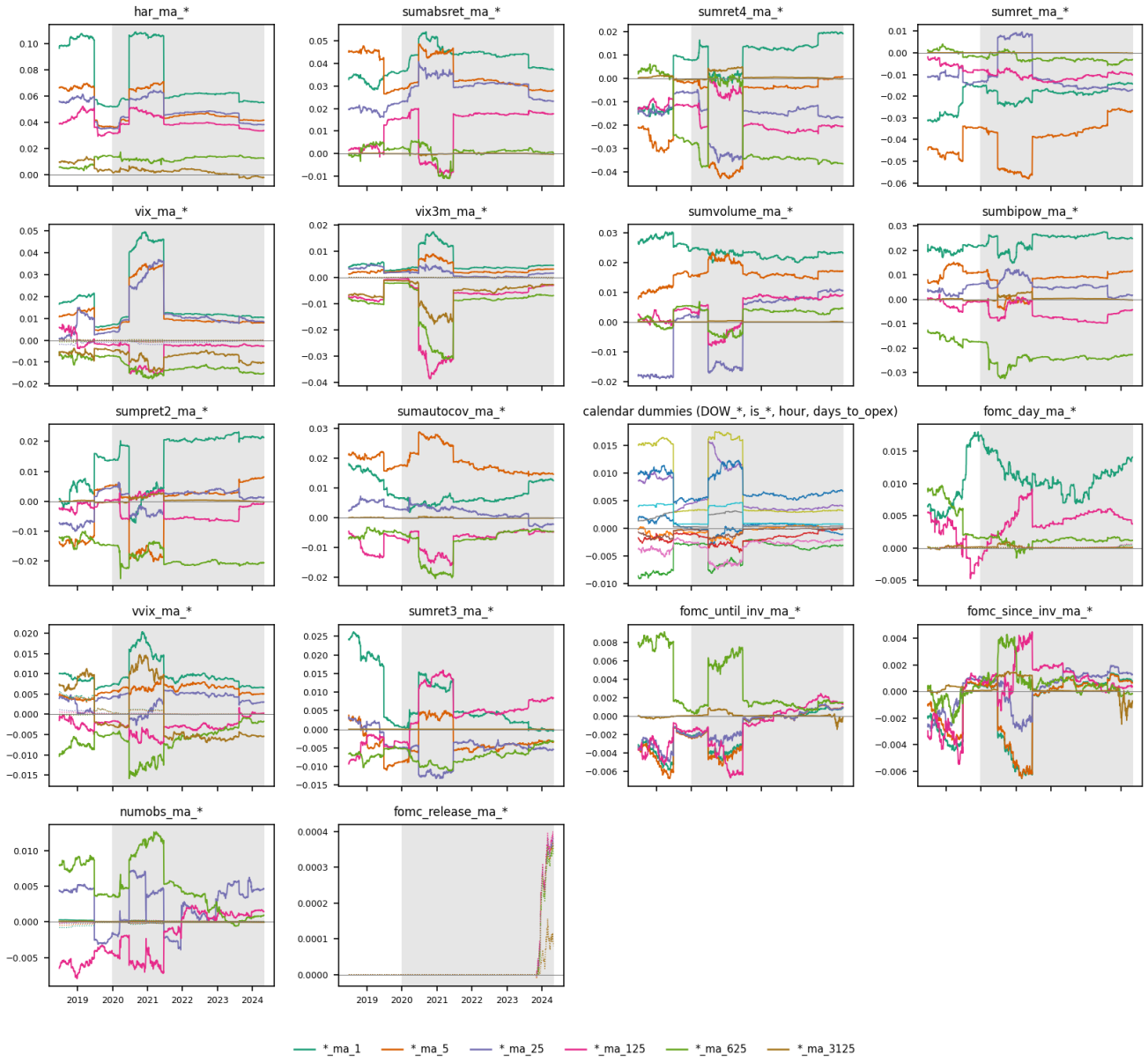


Figure 1: Standardized weights over every refit (1,469 sessions), one panel per input series; solid = the series' rolling means (colour = window length in 30-minute bars). Panel titles are the design's column names; `*_ma_`k is the mean over the last k 30-minute bars (`_ma_1` = the last bar), and the dotted lines are the `*_avail_ma_`k / `*_active_ma_`k is-present / is-nonzero flags. Every exogenous input is divided by its time-of-day baseline (after its transform, e.g. log for the VIX), and `har_ma_*` are trailing means of the adjusted target, so no name carries a prefix. Key: `har_ma` realized variance (the target); `sumret` signed return; `sumabsret` absolute return; `sumret3` cubed returns (skew); `sumret4` 4th-power returns (tails); `sumpret2` upside squared returns; `sumbipow` bipower variation; `sumautocov` return autocovariance; `sumvolume` ES volume (contracts); `numobs` ES trade count per bar; `vix`, `vvix`, `vix3m` the Cboe indices; `fomc_release` FOMC statement bar; `fomc_day` FOMC day; `fomc_until_inv` / `fomc_since_inv` 1 / days to the next / since the last FOMC. Grey band = the 866 deck days. 138 of the 244 design columns carry a nonzero weight on deck days; the rest are constant or duplicated within the window and are masked to zero.

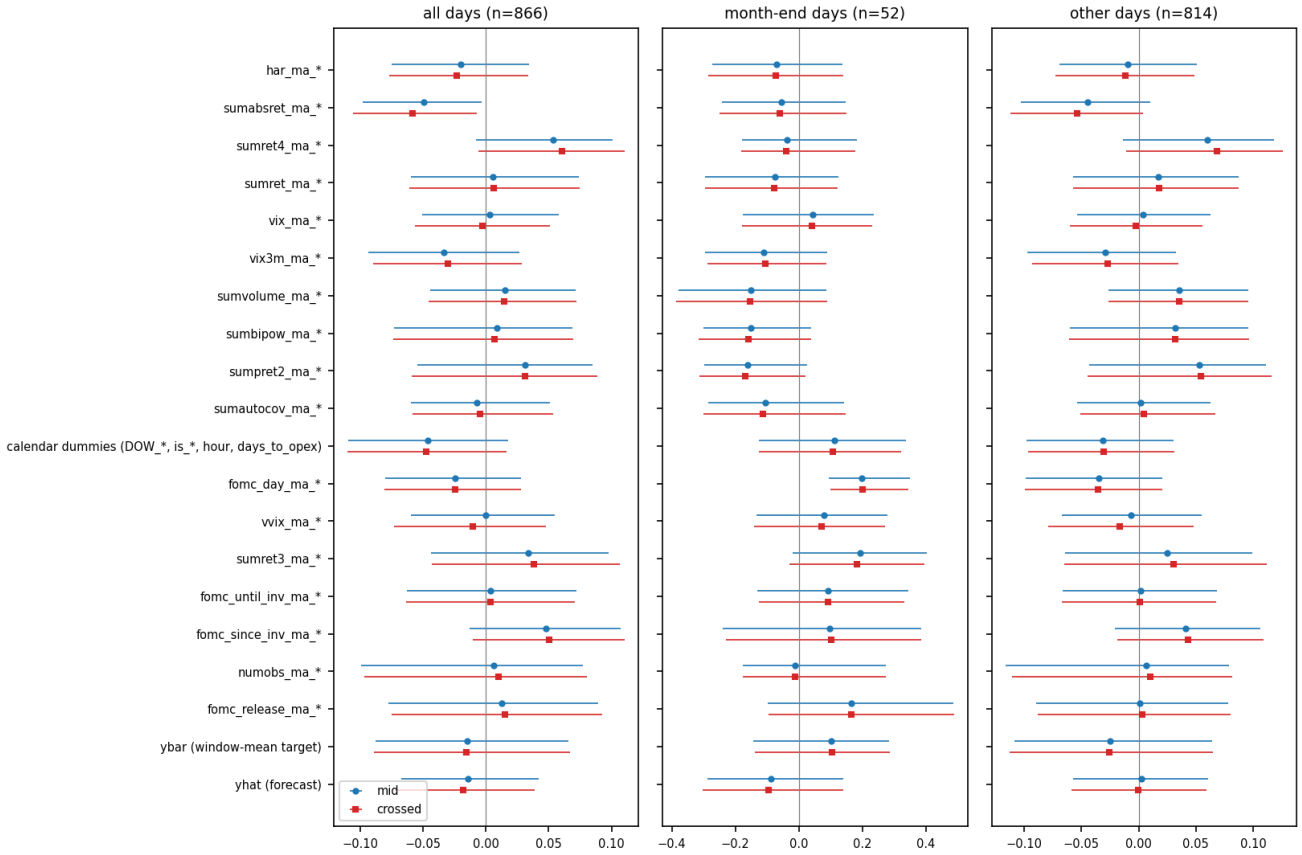


Figure 2: Correlation across deck days of each input series' summed contribution (rows named by the design's columns, key under Figure 1; `ybar` = the level, `yhat` = the forecast) to the forecast with the day's return per premium (blue mid, red crossed), 95% block-bootstrap intervals.

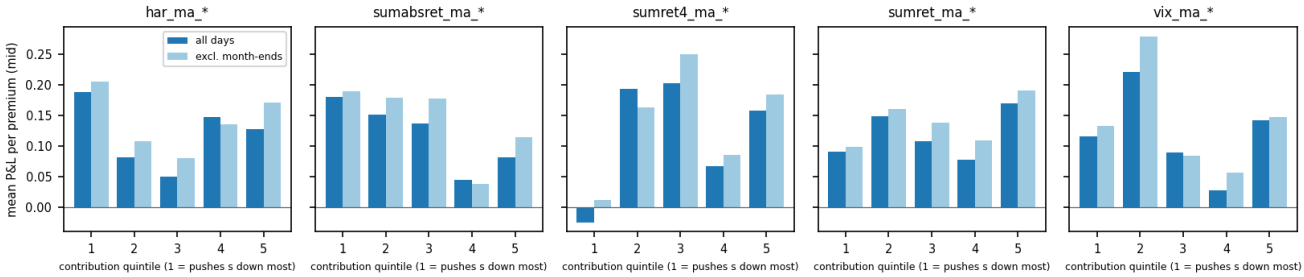


Figure 3: Mean mid return per premium by quintile of each of the five largest series' contributions (named by the design's columns, key under Figure 1) (dark: all 866 days; light: without month-ends).

Regime changes, which inputs dominate, and how the lasso picks out the VIX

same forecast, same 16:00 bar; 1,469 refits (2018-06-25 .. 2024-04-30), of which 866 are deck days

1. Regime changes. *Question:* do the weights, and the trade's money, change at the known dates – the penalty switches (2019-06-26, 2020-06-23, 2021-06-21), the Feb–Apr 2020 crash (2020-02-20 .. 2020-04-30, from the session after the S&P high on the deck, 2020-02-19), the start of every-day expirations – and is a change the penalty or the data?

Gate. Two more re-runs of the same arm with the penalty held at one value: held at 100 it equals the arm of record on all 500 refits where the record used 100 (to 2.6×10^{-12}), held at 1000 on all 969 refits at 1000 (to 9.6×10^{-14}). The arm of record is the two held-penalty runs spliced at the switch dates, so a change that neither held run shows is the penalty. *Every-day expirations.* The first Tuesday/Thursday deck day is 2020-01-21 (a holiday-shifted expiry); every Tuesday session is a deck day from 2022-04-26 and every Thursday from 2022-05-19; the tests use 2022-04-26. *Tests.* Known date: Chow test in Wald form with Newey–West covariance (lag $\lfloor 4(n/100)^{2/9} \rfloor = 7$ on refits, 6 on deck days), $\chi^2(p)$, run only where each side holds at least 15% of the sample; the Feb–Apr 2020 episode (50 refits, 32 deck days) by Chow's predictive F test, the standard test for a stretch too short to fit on its own. Unknown date: Andrews' sup-Wald over the middle 70% of the sample, critical values simulated from its limit (20,000 draws; $p = 1$, 5%: 8.73, published 8.58–8.85). Regressions: the 16:00 target on the forecast (calibration, $p = 2$); the target on the contributions of the six largest series plus the rest summed ($p = 8$); the day's return per premium on a constant ($p = 1$), and on the same contributions ($p = 8$).

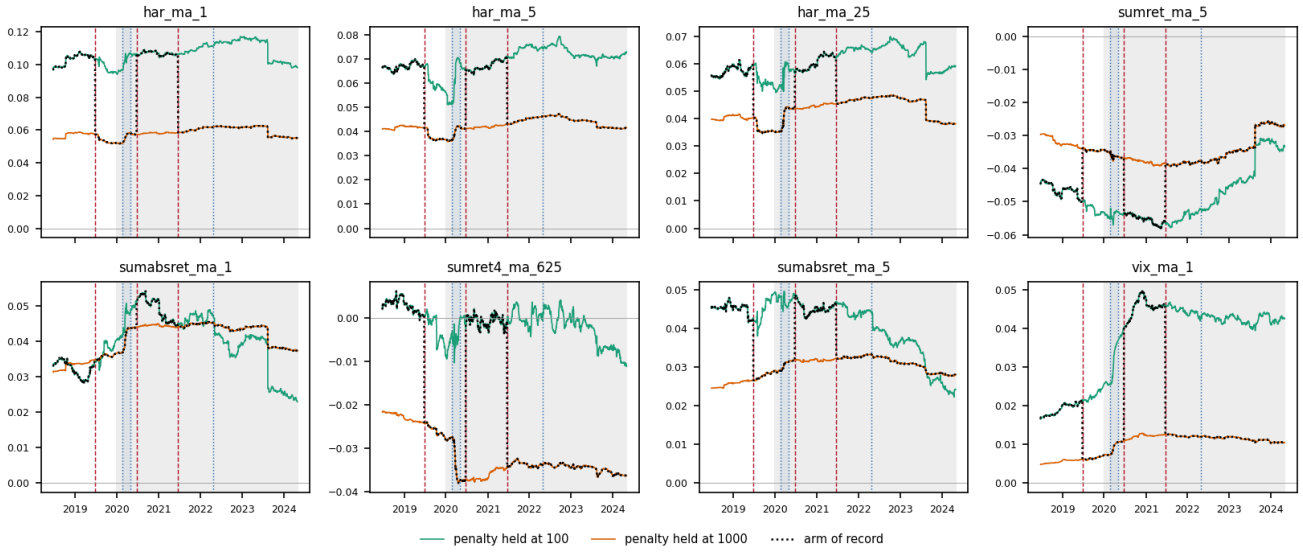


Figure 4: Standardized weights of the eight largest inputs (design column names, key under Figure 1): the arm of record (dotted black) and the same arm with the penalty held at 100 (green) or 1000 (orange). Red dashed = penalty switches; blue dotted = COVID start/end and every-day expirations; blue band = Feb–Apr 2020; grey = deck days.

Table 4: The step in all standardized weights at each penalty switch (Euclidean norm across inputs), split into the penalty (the two held runs on the same day) and one more day of data (the old penalty's run, day over day).

date	penalty	step	penalty part	data part	largest step, other days	median
2019-06-26	100 to 1000	0.0932	0.0930	0.0010	0.0199	0.00062
2020-06-23	1000 to 100	0.1177	0.1177	0.0008	0.0199	0.00062
2021-06-21	100 to 1000	0.1232	0.1235	0.0022	0.0199	0.00062

Reading.

- *Every step in the weights is the penalty.* At the three switches all weights together jump by 0.093, 0.118, 0.123, of which the penalty is 0.093, 0.118, 0.123; on every other refit the largest step is 0.020 and the median 0.0006. Held at one penalty the weights drift smoothly through COVID and through every-day expirations (Figure 4).

Table 5: Break tests: p-values at the known dates and for the COVID episode, and the unknown-date sup-Wald (statistic, date of its maximum, p). record = the arm of record; 100 / 1000 = penalty held. – = not testable (before the deck, or under 15% of the deck days on one side). * = $p < 0.05$; with seven dates per row, a stray * is expected by chance.

regression	pen.	p	pen. 100 to 1000 2019-06-26	COVID start 2020-02-20	COVID end 2020-05-01	pen. 1000 to 100 2020-06-23	pen. 100 to 1000 2021-06-21	every-day 2022-04-26	COVID episode	sup-Wald
target on forecast	record	2	0.713	0.867	0.191	0.119	0.034*	0.040*	<0.001*	7.4 (2021-06-02) 0.262
	100	2	0.672	0.724	0.623	0.435	0.599	0.589	<0.001*	2.6 (2020-09-29) 0.931
	1000	2	0.727	0.975	0.194	0.105	0.210	0.204	<0.001*	5.4 (2020-07-14) 0.515
target on contributions	record	8	0.002*	0.159	0.119	0.047*	0.232	0.022*	<0.001*	29.1 (2022-10-11) 0.007*
	100	8	0.023*	0.181	0.103	0.149	0.190	0.280	<0.001*	28.9 (2020-04-09) 0.008*
	1000	8	0.464	0.648	0.006*	0.021*	0.076	0.008*	<0.001*	40.0 (2022-10-11) <0.001*
mean return, mid	record	1	–	–	–	–	0.791	0.779	0.768	2.2 (2023-01-04) 0.752
mean return, crossed	record	1	–	–	–	–	0.962	0.569	0.679	2.8 (2023-01-04) 0.615
return on contributions	record	8	–	–	–	–	0.011*	0.003*	0.600	25.4 (2021-06-09) 0.028*
	1000	8	–	–	–	–	0.007*	0.004*	0.537	27.0 (2021-02-12) 0.017*

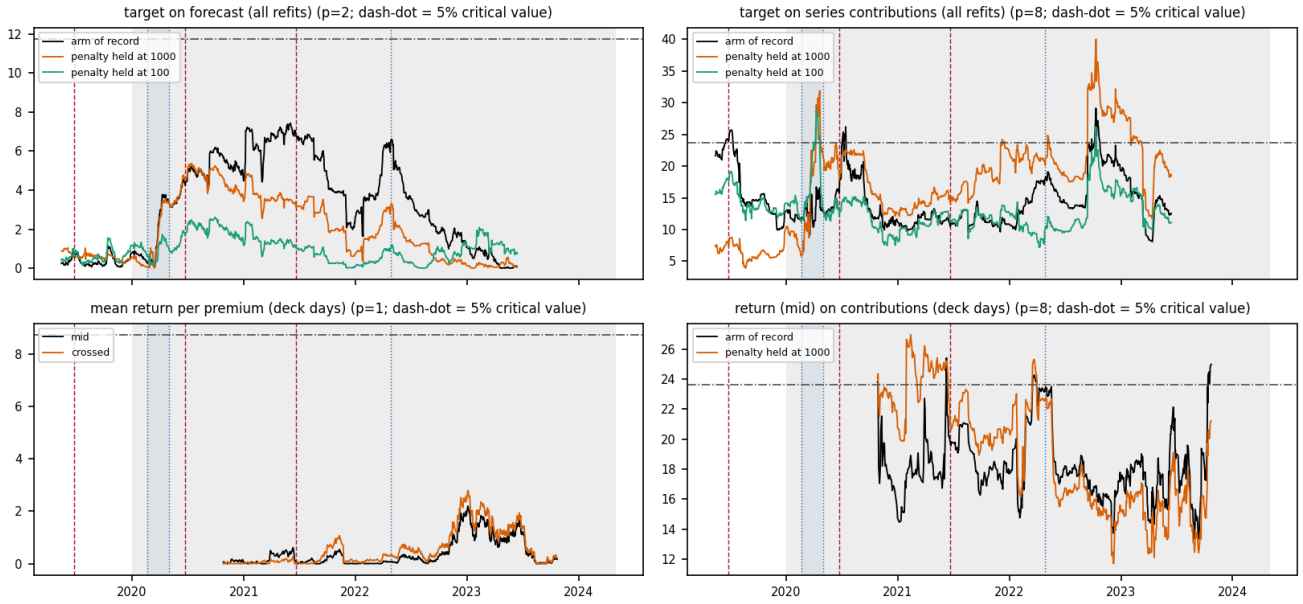


Figure 5: Wald statistic for a break at each date (the sup-Wald scan) with its 5% critical value. Vertical lines as in Figure 4.

- *The forecast's calibration does not break.* Target on forecast: sup-Wald 7.43 (p 0.26); held at 100 / 1000: 2.61 / 5.37. The two known-date rejections in the arm of record (2021-06-21, p 0.034; 2022-04-26, p 0.040) vanish with the penalty held ($p \geq 0.20$): made by the penalty. The Feb–Apr 2020 episode rejects under every penalty ($p < 0.001$); the predictive test reacts to the crash's larger misses as well as to a changed slope.
- *How the series' contributions map into realized variance does change, and not because of the penalty.* The sup-Wald rejects under each penalty (29.1, 28.9, 40.0; $p \leq 0.008$), its maximum at 2022-10-11 (record, held 1000) or 2020-04-09 (held 100): one date is not pinned down. The COVID episode differs under every penalty. At the known dates the answer depends on which penalty is held (e.g. 2022-04-26: record 0.022, held 100 0.28, held 1000 0.008), so none is a clean data break.
- *The trade's money has no break in level.* Mean return per premium: sup-Wald 2.19 mid / 2.79 crossed (p 0.75 / 0.61); the COVID days' mean is not different (p 0.77 / 0.68). The return's slopes on the contributions do shift (sup-Wald 25.4, p 0.028, at 2021-06-09; held 1000: 27.0, p 0.017) – the data, not the penalty – but these are eight slopes on heavy-tailed returns that are near zero on either side (Table 2).
- *Not testable on the money:* 2019-06-26 is before the deck; the COVID start/end and 2020-06-23 leave under 15% of the deck days on one side.

2. Which series dominate (rolling SHAP). *Question:* which input series carry the forecast, and does that change across regimes?

Why the contributions are the SHAP values. For a linear forecast $\hat{y} = \beta_0 + \sum_j \beta_j x_j$ with the window mean $\bar{x}$ as the baseline (interventional SHAP), the SHAP value of input j is exactly $\beta_j(x_j - \bar{x}_j)$, and the values sum to $\hat{y}$ minus the forecast at $\bar{x}$. *Gate.* The `shap` library's LinearExplainer (the window mean as its one background row) on 8 refits (the first, the last, and each candidate date) equals $\beta_j(x_j - \bar{x}_j)$ to 0, and its values plus the baseline equal the forecast to 1.6×10^{-16} (relative). A series' SHAP value is the sum over its columns (all lags and flags). Share = the series' mean |SHAP| over a trailing window divided by the sum over series. The window is a choice: 63 sessions (a quarter), checked against 126 (a half-year).

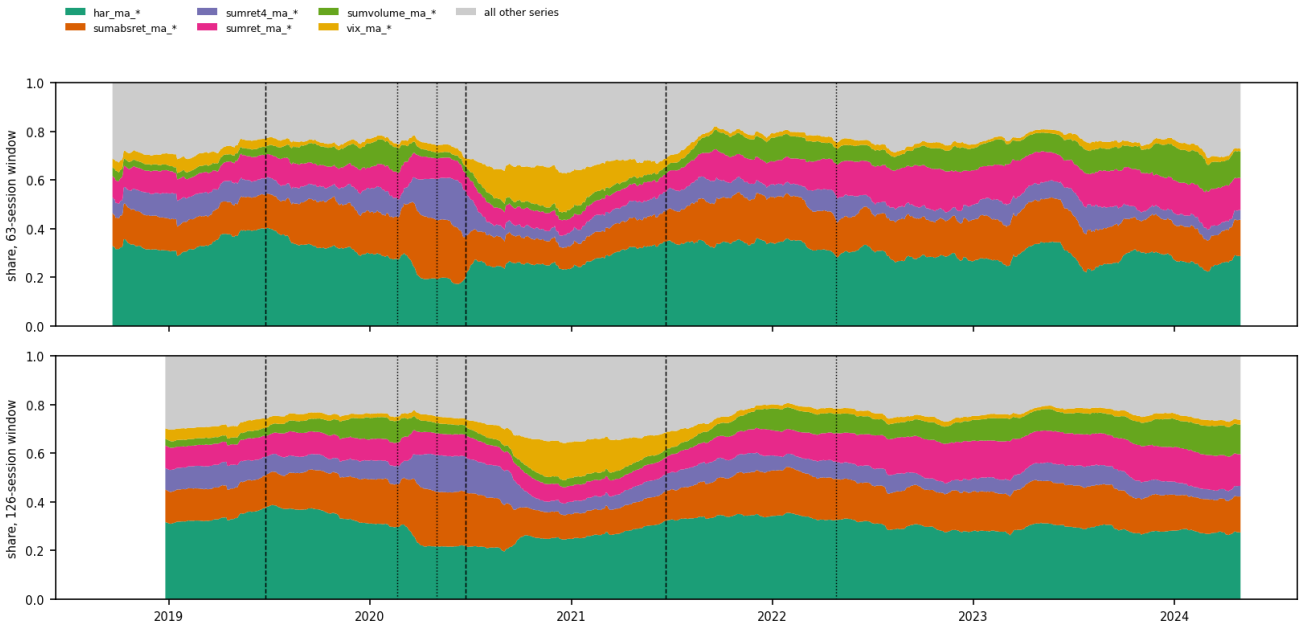


Figure 6: Share of mean |SHAP| by input series (design column names, key under Figure 1), trailing 63 sessions (top) and 126 sessions (bottom). Dashed = penalty switches; dotted = COVID start/end and every-day expirations.

Table 6: The three largest series by share of mean |SHAP| within each regime (cut at the candidate dates; no rolling), and the second and third with the penalty held at 1000.

regime	refits	deck days	penalty	first	second	third	held 1000: second / third
2018-06-25 .. 2019-06-25	250	0	100	`har_ma_*` 35%	`sumabsret_ma_*` 13%	`sumret_ma_*` 9%	`sumabsret_ma_*` / `sumret_ma_*`
2019-06-26 .. 2020-02-19	164	21	1000	`har_ma_*` 30%	`sumabsret_ma_*` 18%	`sumret_ma_*` 10%	`sumabsret_ma_*` / `sumret_ma_*`
2020-02-20 .. 2020-04-30	50	32	1000	`sumabsret_ma_*` 25%	`har_ma_*` 19%	`sumret4_ma_*` 18%	`har_ma_*` / `sumret4_ma_*`
2020-05-01 .. 2020-06-22	36	23	1000	`har_ma_*` 29%	`sumabsret_ma_*` 16%	`sumret_ma_*` 9%	`sumabsret_ma_*` / `sumret_ma_*`
2020-06-23 .. 2021-06-18	250	154	100	`har_ma_*` 28%	`vix3m_ma_*` 14%	`vix_ma_*` 12%	`sumabsret_ma_*` / `sumret_ma_*`
2021-06-21 .. 2022-04-25	213	136	1000	`har_ma_*` 33%	`sumabsret_ma_*` 18%	`sumret_ma_*` 12%	`sumabsret_ma_*` / `sumret_ma_*`
2022-04-26 .. 2024-04-30	506	500	1000	`har_ma_*` 29%	`sumabsret_ma_*` 16%	`sumret_ma_*` 14%	`sumabsret_ma_*` / `sumret_ma_*`

Reading.

- *Realized variance dominates almost everywhere.* `har_ma_*` has the largest share on 95% of refits (63-session window) and 95% (126); absolute return (`sumabsret_ma_*`) leads on the rest. The two windows name the same leader on 98.6% of 1,344 refits.
- *COVID.* In Feb–Apr 2020 absolute return leads (25%) ahead of realized variance (19%) and 4th-power returns (18%) – the only regime where realized variance is not first.
- *The penalty-100 year* (2020-06-23 .. 2021-06-18) puts VIX3M and VIX second and third (14%, 12%); with the penalty held at 1000 the same year ranks absolute and signed returns second and third, like every other regime: the VIX's rise there is the penalty.
- *Every-day expirations* (from 2022-04-26) leave the ranking unchanged.

3. The lasso: how the VIX is picked out. *Question:* when the same forecast is fit by the spec's recursive lasso instead of the ridge, which VIX columns does it keep, with what sign and size, and do they enter or leave at the regime dates?

Gate. The lasso re-run (same bucket, bar, 2000-session window) equals the stored lasso forecast on all 1,469 rows of the 16:00 bar (max relative difference 5.3×10^{-13}); coefficients · inputs + intercept reproduce it to 5.0×10^{-16}. Its penalty (grid 10^{-6} .. 10^{-2}, re-chosen every 250 refits) is 0.01 (the top of the grid) until 2021-06-21 and 0.001 after. A second re-run with the penalty held at 0.001 equals the lasso of record on its 719 refits at that penalty (to 0). The lasso keeps 12 to 56 of the 244 columns (median 16); its forecast correlates 0.986 with the ridge's.

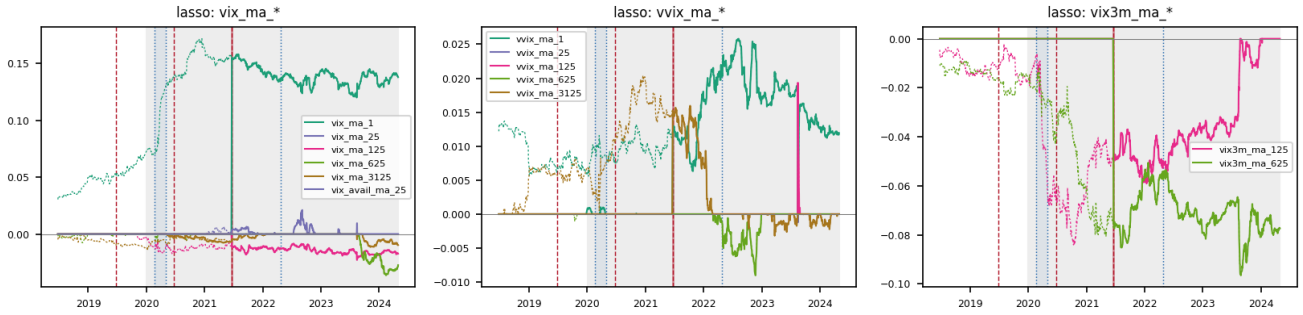


Figure 7: The lasso's standardized weights on the VIX-family columns (design column names; colour = lag as in Figure 1): solid = lasso of record, dashed = penalty held at 0.001 from the start. Thick red line = the lasso's penalty re-choice; other lines as in Figure 4. Only columns the lasso of record ever keeps are drawn.

Table 7: VIX-family columns in the lasso: refits (of 1,469) and deck days with a nonzero weight, share positive, mean standardized weight when nonzero, first nonzero refit, number of entries and exits; with the penalty held at 0.001: refits nonzero before 2021-06-21 (of 750) and the mean weight there.

column	refits	deck days	positive	mean std. weight	first	entries/exits	held: refits before	held: mean
vix_ma_1	719	636	100%	+0.1388	2021-06-21	1	750	+0.0923
vix_ma_25	215	174	100%	+0.0052	2021-06-21	18	100	+0.0030
vix_ma_125	719	636	0%	-0.0143	2021-06-21	1	414	-0.0109
vix_ma_625	170	169	0%	-0.0245	2023-08-24	3	197	-0.0022
vix_ma_3125	568	415	0%	-0.0046	2020-05-07	15	741	-0.0057
vvix_ma_1	767	664	100%	+0.0155	2019-12-27	11	750	+0.0092
vvix_ma_25	1	1	0%	-0.0012	2023-08-16	2	0	–
vvix_ma_125	9	9	100%	+0.0089	2023-08-15	4	0	–
vvix_ma_625	205	190	0%	-0.0035	2022-03-02	6	15	-0.0005
vvix_ma_3125	366	301	47%	+0.0043	2021-06-21	32	698	+0.0089
vix3m_ma_125	634	551	0%	-0.0376	2021-06-21	6	750	-0.0309
vix3m_ma_625	719	636	0%	-0.0714	2021-06-21	1	750	-0.0302
vix_avail_ma_25	5	5	0%	-0.0001	2022-08-10	4	0	–
vvix_avail_ma_1	0	0	–	–	–	0	416	+0.0061

Reading.

- *The VIX enters at the penalty re-choice, not at an event.* Before 2021-06-21 no VIX-family column in the lasso of record exceeds 0.0076 in standardized weight. On 2021-06-21, when the penalty drops to 0.001, `vix_ma_1` enters and stays on all 719 later refits, always positive (mean +0.139), with the longer VIX3M means negative (`vix3m_ma_625` -0.071, `vix3m_ma_125` -0.038): the VIX's last bar against the 3-month index's longer averages.
- *The VIX was in the data before.* Held at 0.001 from the start, the lasso keeps `vix_ma_1` on 750 of the 750 earlier refits (mean +0.092) and `vix3m_ma_625` negative on 750, through COVID. The top-of-grid penalty kept the VIX out until 2021.
- *Entries and exits in between are small.* Of the 104 VIX-family entries and exits, 7 are at the penalty re-choice and 97 in the day-to-day path; after a day-to-day entry no weight exceeds 0.0177. None marks COVID or every-day expirations.
- *The lasso leans on the VIX more than the ridge.* Share of mean |SHAP| on deck days: VIX 9.8% and VIX3M 8.8% (ridge 4.8% and 5.1%); realized variance 44.1% (ridge 28.2%).

Files: `experiments/model_diagnostics_1530_regimes.py`; outputs `results/model_diagnostics_1530/regimes_*`.

Feature importance: tree models against the linear models

same 16:00 bar, same 244 design columns (live-feasible), same 1,469 forecasts (2018-06-25 .. 2024-04-30), of which 866 are deck days

Question. Which input series carry the tree forecasts (LightGBM, XGBoost, random forest), and are they the series that carry the per-bar ridge and lasso?

The models. Each tree model is fit on exactly the per-bar linear design (same columns, same target, same 2000-session window) and refit every 10 sessions; LightGBM and XGBoost keep the hyperparameters of an earlier tuning on the pooled design (minimum leaf size scaled to the 2000-row window), the random forest uses the library defaults (100 trees). On this bar their QLIKE against the ridge's, on the same 1,406 forecasts (difference in %, day-block 95% interval): LightGBM -1.0% [-8.3, +5.5], XGBoost +0.8% [-6.9, +7.8], random forest -0.2% [-8.1, +6.6] – every interval covers zero. The 15:30 sign(s) trade on their forecasts: Sharpe mid (crossed) LightGBM 1.69 (1.22), XGBoost 1.70 (1.23), random forest 1.58 (1.11); on the ridge's forecast 1.90 (1.44), same 866 days.

Importance = share of mean |SHAP|. For every forecast, the SHAP value of an input is its part of (forecast − base value); a series' SHAP value is the sum over its columns (all lags, the is-present / is-nonzero flags, the HAR ladder's open/close interactions), and its share is its mean |SHAP| divided by the sum over the 18 series – the measure of Section 2. Linear models: $\beta_j(x_j - \bar{x}_j)$ with the window mean as baseline (checked against the `shap` library in Section 2). Trees: the `shap` library's TreeExplainer on the model in force for that forecast; base value = the explainer's expected value (the mean prediction over the training window). Both are in the forecast's units, so the shares are comparable across models.

Gate. Every model's SHAP values plus its base value equal its forecast on all 1,469 rows (max relative difference: ridge 1.2×10^{-15}, lasso 5.5×10^{-16}, LightGBM 7.9×10^{-8}, XGBoost 1.6×10^{-6}, random forest 7.6×10^{-8}; the trees' values are stored in single precision). The trees' forecasts in the SHAP files equal their scored forecasts exactly, all five models cover the same stamps and columns, and the ridge and lasso shares reproduce Sections 2–3 exactly.

Table 8: Share of mean |SHAP| by input series (%), 16:00 bar, live-feasible design. Series named by the design's columns (key under Figure 1). † `har_ma_*`, `sumabsret_ma_*`, `sumret4_ma_*`, `sumbipow_ma_*`, `sumpret2_ma_*` (the series that measure the size of recent returns) summed before taking $|\cdot|$: their joint share.

	all 1,469 forecasts					866 deck days				
series	ridge	lasso	LGBM	XGB	RF	ridge	lasso	LGBM	XGB	RF
`har_ma_*`	29.6	50.6	57.2	64.1	66.1	28.2	44.1	56.2	62.7	64.0
`sumret_ma_*`	9.9	10.4	9.9	9.6	6.7	10.9	9.5	10.2	9.7	7.0
`sumabsret_ma_*`	15.4	6.6	3.1	4.8	6.4	15.7	5.6	3.2	5.0	7.3
`sumvolume_ma_*`	5.4	3.4	5.7	6.6	4.6	5.9	3.0	6.9	8.0	5.6
`sumret4_ma_*`	7.9	4.3	2.6	1.5	1.3	7.6	4.8	2.3	1.4	1.2
`vix_ma_*`	4.8	7.1	2.1	1.1	1.1	4.8	9.8	1.8	1.0	1.1
`sumautocov_ma_*`	4.1	3.5	3.3	2.3	1.9	3.9	3.4	3.0	2.1	1.8
calendar (DOW_*, is_*, hour, days_to_opex)	2.6	2.7	3.9	2.9	2.2	2.8	4.0	4.4	3.2	2.4
`vix3m_ma_*`	4.5	6.5	1.0	0.6	0.8	5.1	8.8	1.0	0.6	0.8
`sumret3_ma_*`	2.2	1.0	2.2	1.5	1.8	1.9	1.4	2.3	1.5	1.8
`sumbipow_ma_*`	4.0	0.8	1.2	1.0	0.9	4.0	1.0	1.2	1.0	0.9
`sumpret2_ma_*`	3.5	0.2	1.5	0.7	1.4	3.3	0.3	1.4	0.6	1.3
`vvix_ma_*`	2.1	1.6	1.7	0.8	1.1	2.3	2.3	1.7	0.8	1.2
`fomc_since_inv_ma_*`	1.0	0.3	1.4	0.8	1.0	0.9	0.4	1.4	0.7	1.1
`fomc_until_inv_ma_*`	1.2	0.2	1.2	0.7	1.1	0.9	0.3	1.2	0.7	1.1
`numobs_ma_*`	0.3	0.1	1.4	0.8	1.3	0.3	0.1	1.1	0.7	1.2
`fomc_day_ma_*`	1.3	0.8	0.4	0.3	0.2	1.4	1.2	0.5	0.2	0.2
`fomc_release_ma_*`	0.1	0.0	0.1	0.1	0.1	0.2	0.0	0.1	0.1	0.1
return-size series summed†	52.7	59.4	63.1	70.8	74.3	52.1	51.6	61.8	69.3	72.7

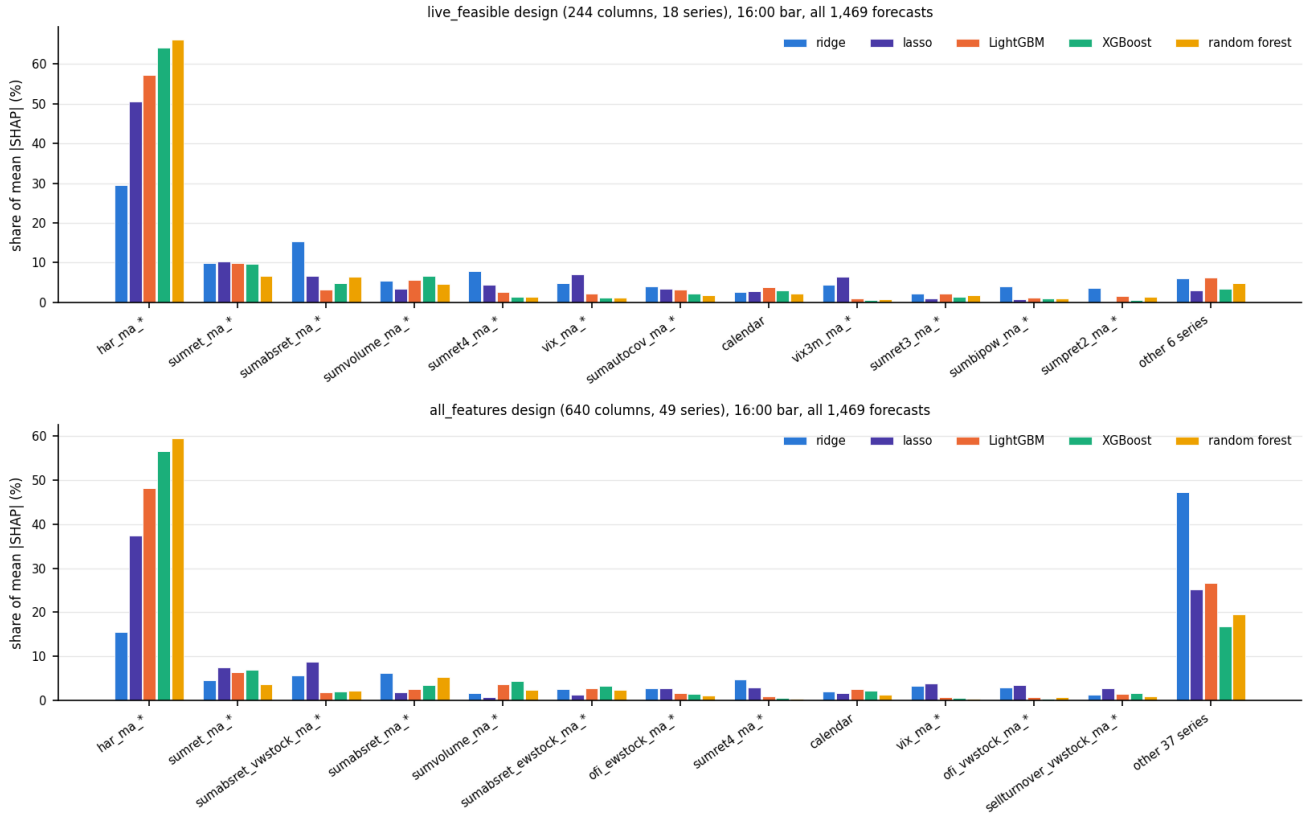


Figure 8: Share of mean |SHAP| by series, one bar per model (all forecasts of the 16:00 bar). Top: live-feasible design (Table above). Bottom: the all-features design (640 columns, 49 series), the 12 largest series and the rest summed. LGBM = LightGBM, XGB = XGBoost, RF = random forest.

Table 9: Rank agreement between models: Spearman correlation of the 18-series share vectors, and how many of each model's five largest series the other shares.

	Spearman		top-5 shared	
models	all	deck days	all	deck days
ridge – lasso	0.89	0.84	3	3
ridge – LightGBM	0.71	0.71	3	4
ridge – XGBoost	0.72	0.72	4	4
ridge – random forest	0.64	0.57	4	4
lasso – LightGBM	0.68	0.67	2	3
lasso – XGBoost	0.69	0.68	3	3
lasso – random forest	0.56	0.46	3	3
LightGBM – XGBoost	0.94	0.94	4	5
LightGBM – random forest	0.95	0.90	4	5
XGBoost – random forest	0.89	0.84	5	5

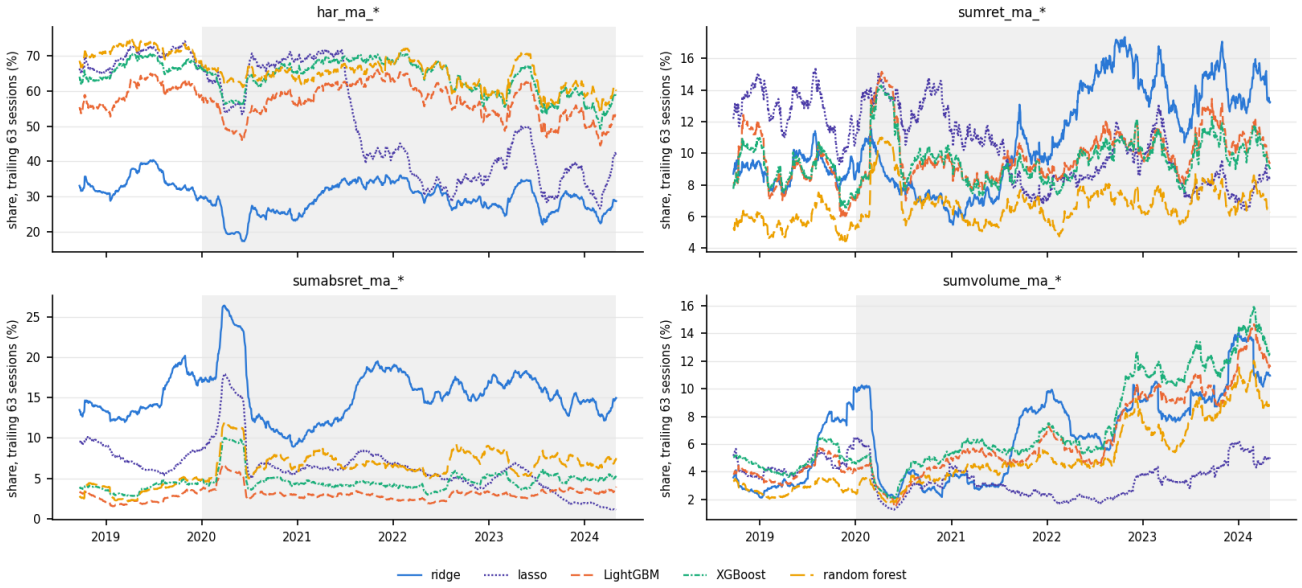


Figure 9: Share of mean |SHAP| over a trailing 63-session window for the four largest series, every model (the trees' SHAP values are per forecast, so the share can be followed over time like the ridge's in Section 2). Grey = the deck period.

Table 10: All-features design (640 columns, 49 series; adds equal- and value-weighted constituent-stock return moments, turnover, spreads and order-flow imbalance, StockTwits message counts and sentiment, and Cboe option volume demand to the live-feasible series): share of mean |SHAP| (%), all 1,469 forecasts of the 16:00 bar. Gate: SHAP plus base value equal the forecast to ridge 7.1×10^{-14}, lasso 6.9×10^{-16}, LightGBM 8.2×10^{-8}, XGBoost 1.8×10^{-6}, random forest 7.5×10^{-8} (relative); the ridge and lasso coefficient re-runs equal their stored forecasts to 3.5×10^{-11} / 2.4×10^{-13}. Spearman over the 49 series: tree–tree 0.81–0.88; ridge–lasso 0.83; tree–linear 0.26–0.51.

series	ridge	lasso	LGBM	XGB	RF
`har_ma_*`	15.4	37.4	48.2	56.4	59.5
`sumret_ma_*`	4.5	7.5	6.3	7.0	3.7
`sumabsret_vwstock_ma_*`	5.7	8.7	1.8	2.1	2.3
`sumabsret_ma_*`	6.3	1.9	2.5	3.4	5.4
`sumvolume_ma_*`	1.6	0.8	3.7	4.5	2.4
`sumabsret_ewstock_ma_*`	2.5	1.3	2.8	3.2	2.4
`ofi_ewstock_ma_*`	2.7	2.7	1.7	1.4	1.2
`sumret4_ma_*`	4.7	3.0	0.9	0.6	0.4
calendar (DOW_*, is_*, hour, days_to_opex)	2.0	1.6	2.6	2.1	1.2
`vix_ma_*`	3.3	3.8	0.7	0.5	0.5
`ofi_vwstock_ma_*`	2.9	3.4	0.7	0.4	0.7
`sellturnover_vwstock_ma_*`	1.2	2.8	1.4	1.6	0.9
other 37 series	47.2	25.3	26.6	16.7	19.5

Reading.

- *Realized variance leads for every model; the trees lean on it far more.* `har_ma_*` carries 29.6% of the ridge's mean |SHAP|, 50.6% of the lasso's and 57.2–66.1% of the trees'. It is the leading series on 95%, 100%, 100%, 100%, 100% of the trailing-63 windows (ridge, lasso, LightGBM, XGBoost, random forest). Over time the ridge's share stays between 17.3 and 40.3%, the trees' between 44.5 and 75.3%; the lasso's share is 66.9% before its penalty re-choice (2021-06-21, Section 3) and 37.5% after.
- *The ridge spreads the size of recent returns over several correlated series* (each return-size column correlates 0.44–0.86 with realized variance at the same lag on these forecasts). Absolute return has 15.4% in the ridge against 3.1–6.4% in the trees, 4th-power returns 7.9% against 1.3–2.6%. Summed into one series with realized variance, bipower and upside squared returns, the return-size family carries 52.7% of the ridge and 63.1–74.3% of the trees. Counting the family as one series closes 41–63% of the tree–ridge gap in realized variance: that part is the ridge splitting one signal across correlated columns (some with negative weights, Figure 1); the rest is the trees using the size of returns more.

- *The VIX matters to the linear models, hardly to the trees.* VIX / VIX3M shares: ridge 4.8 / 4.5%, lasso 7.1 / 6.5%, trees at most 2.1 / 1.0% (the linear models weigh the VIX against the 3-month index with opposite signs, Section 3). Signed return ranks 3, 2, 2, 2, 2 (ridge, lasso, LightGBM, XGBoost, random forest).
- *Rank agreement.* The three trees agree with one another (Spearman 0.89–0.95), ridge and lasso with each other (0.89), trees with the linear models less (0.56–0.72); all five name the same leading series. The trees' own split-gain / impurity importance ranks the series like their SHAP (Spearman 0.89, 0.97, 0.86) and also puts realized variance first (61.7%, 73.6%, 67.1%).
- *All features.* With 640 columns the trees still put 48.2–59.5% on realized variance; no added series exceeds 3.2% in any tree. The linear models spread further: realized variance 15.4% (ridge) and 37.4% (lasso), the return-size family 21.5% and 39.6% (trees 51.3–64.9%); their largest added series is `sumabsret_vwstock_ma_*` (5.7% ridge, 8.7% lasso). Tree and linear rankings agree less than on the live-feasible design (Spearman 0.26–0.51); every model still leads with realized variance.

Files: `experiments/model_diagnostics_1530_trees.py`; outputs `results/model_diagnostics_1530/trees_shap*`.